



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

CHEMISTRY
PAPER 1

9189/1

JUNE 2017 SESSION

2 hours

Additional materials:

Answer paper

Data Booklet

Mathematical tables and/or electronic calculator

TIME: 2 hours

INSTRUCTIONS TO CANDIDATES

Write your name, Centre number and candidate number in the spaces provided on the answer paper/answer booklet.

Answer **six** questions.

Answer **two** questions from Section A, **one** question from Section B, **two** questions from Section C and **one** other question chosen from any section.

Write your answers on the separate answer paper provided.

If you use more than one sheet of paper, fasten the sheets together.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

You are reminded of the need for good English and clear presentation in your answers.

This question paper consists of 7 printed pages and 1 blank page.

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Section A

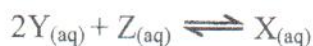
Answer at least **two** questions from this section.

- 1 (a) (i) Define the term *equilibrium constant*.
- (ii) State the factor that affects an equilibrium constant.
- (iii) Explain how the factor in (ii) affects an equilibrium constant.

[3]

- (b) Samples of 1.00 mol of Y and 0.75 mol of Z were mixed and allowed to reach equilibrium in a closed container.

They react according to the equation:



At equilibrium 0.70 mole of Y and 0.60 mol of Z were found to be present in the reaction mixture.

Calculate the equilibrium constant.

[3]

- (c) A titration was carried out using acidified potassium dichromate (VI) in a conical flask and iron (II) sulphate in a burette.
- (i) Write a balanced equation for the reaction taking place.
- (ii) Calculate the E_{cell}^{θ} .
- (iii) Name the indicator used in this titration.
- (iv) Sketch a graph to show how the e.m.f. changes during the titration.

[6]

[Total: 12]

- 2 (a) A gaseous hydrocarbon of volume 10 cm^3 was mixed with 50 cm^3 of excess oxygen and the mixture sparked so that the hydrocarbon was completely burnt. The volume of gas remaining at the end of combustion was 35 cm^3 . After passing over sodalime, this volume was reduced to 5 cm^3 . All gases were measured at 25°C and at the same pressure.

- (i) Calculate the volume of
1. oxygen used,
 2. carbon dioxide produced.
- (ii) Deduce the molecular formula of the hydrocarbon.

[5]

- (b) (i) Define the term *standard enthalpy change of solution*.
- (ii) Describe in terms of energetic processes the dissolution of an ionic compound.
- (iii) Suggest the reasons for the low solubility of MgO in water.

[7]

[Total: 12]

- 3 (a) Hydrogen carbonate ions provide a buffering action in biological systems.

- (i) Define the term *buffer solution*.
- (ii) Explain by means of equations, the buffering action of hydrogen carbonate ions.

[3]

- (b) A mixture of benzoic acid and sodium benzoate can also act as a buffer.

$$[\text{K}_a \text{ of benzoic acid is } 6.4 \times 10^{-5} \text{ mol dm}^{-3}]$$

Calculate the

- (i) concentration of H^+ ions in 0.02 mol dm^{-3} benzoic acid,
- (ii) pH of 0.02 mol dm^{-3} benzoic acid,
- (iii) pH of a solution containing 7.2 g of sodium benzoate in 1 dm^3 of 0.02 mol dm^{-3} benzoic acid,
- (iv) pH change, if 1 cm^3 of 1.0 mol dm^{-3} NaOH is added to the buffer in (iii).

[7]

- (c) Show that at 25°C , the pH of water is 7.

[2]

[Total: 12]

Section B

Answer at least **one** question from this section.

- 4 (a) Differentiate between covalent radius and Van der Waal's radius. [4]
- (b) Magnesium chloride is a high melting point solid, aluminium chloride is a solid which sublimes readily at about 180 °C and silicon tetrachloride is a volatile liquid.

Explain the differences in the thermal volatilities of these chlorides. [3]

- (c) Table 4.1 shows the melting points of Group (IV) elements.

Table 4.1

element	melting point/°C
C(diamond)	3 550
Si	1 410
Ge	937
Sn	232
Pb	327

Explain the trend observed in the melting points of the elements. [5]
[Total: 12]

- 5 (a) (i) State any **two** uses of ceramics.
- (ii) Relate the uses of ceramics to their properties. [5]
- (b) Compare and contrast the following properties of carbon dioxide and silicon dioxide:
- (i) structure and bonding
- (ii) acid/base nature [4]
- (c) Explain why SiO₂ is more stable than SiO whereas PbO is more stable than PbO₂. [3]

[Total: 12]

Section C

Answer at least **two** questions from this section.

- 6 A carboxylic acid, **P**, has the following composition by mass

carbon	-	62.10%
oxygen	-	27.60%
hydrogen	-	10.30%

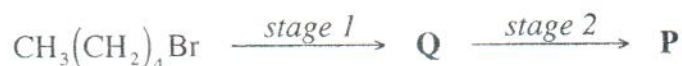
Neutralisation of a sample of 0.10 g of **P** requires 8.60 cm³ of 0.10 mol dm⁻³ sodium hydroxide.

- (a) Calculate the

- (i) empirical formula of **P**,
- (ii) molecular formula of **P**.

[5]

- (b) **P** can be synthesized in the laboratory from 1-bromopentane by a two stage reaction sequence.



- (i) Suggest the reagents and conditions for each stage.
- (ii) Name the reaction that occurs in *stage 2*.
- (iii) Deduce the structural formula of

1. **P**,

2. **Q**.

[7]

[Total: 12]

7 Fig. 7.1 shows the structure of lanosterol.

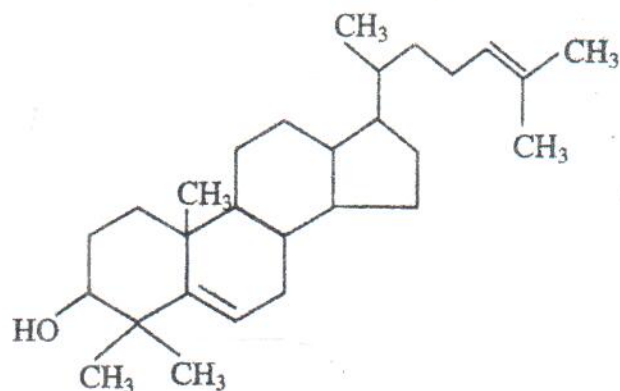


Fig. 7.1

- (a) State the number of chiral centres present in this molecule. [1]
- (b) (i) Name any **two** functional groups present in lanosterol.
- (ii) Describe reactions that can be carried out to confirm the presence of the functional groups named in (b)(i). [6]
- (c) Give the structural formula(e) of the organic product(s) formed when lanosterol reacts with
- (i) hot potassium permanganate under reflux,
- (ii) phosphorus pentachloride,
- (iii) cold potassium permanganate.

[5]

[Total: 12]

- 8 The structures of aspirin and paracetamol are shown in **Fig. 8.1**.

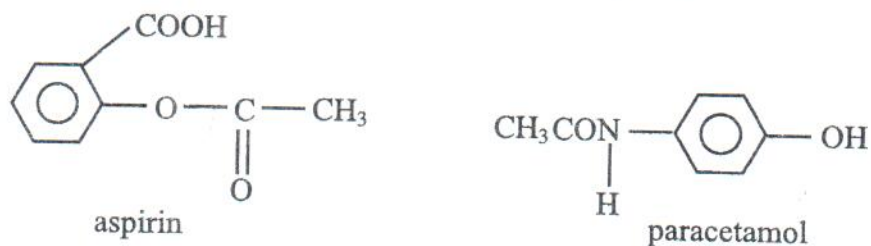


Fig. 8.1

- (a) Name the linkage in
1. paracetamol,
 2. aspirin.
- [2]
- (b) (i) Draw the structural formulae of organic products formed when compounds in **Fig. 8.1** are hydrolysed.
- [5]
- (ii) State the reagent and conditions used in the hydrolysis reaction.
- [2]
- (c) Describe a reaction to distinguish paracetamol from aspirin.
- [2]
- (d) Comment on the solubility of paracetamol in water.
- [2]
- (e) Draw the structural formula of the compound formed when paracetamol reacts with aspirin.
- [1]
- [Total: 12]

